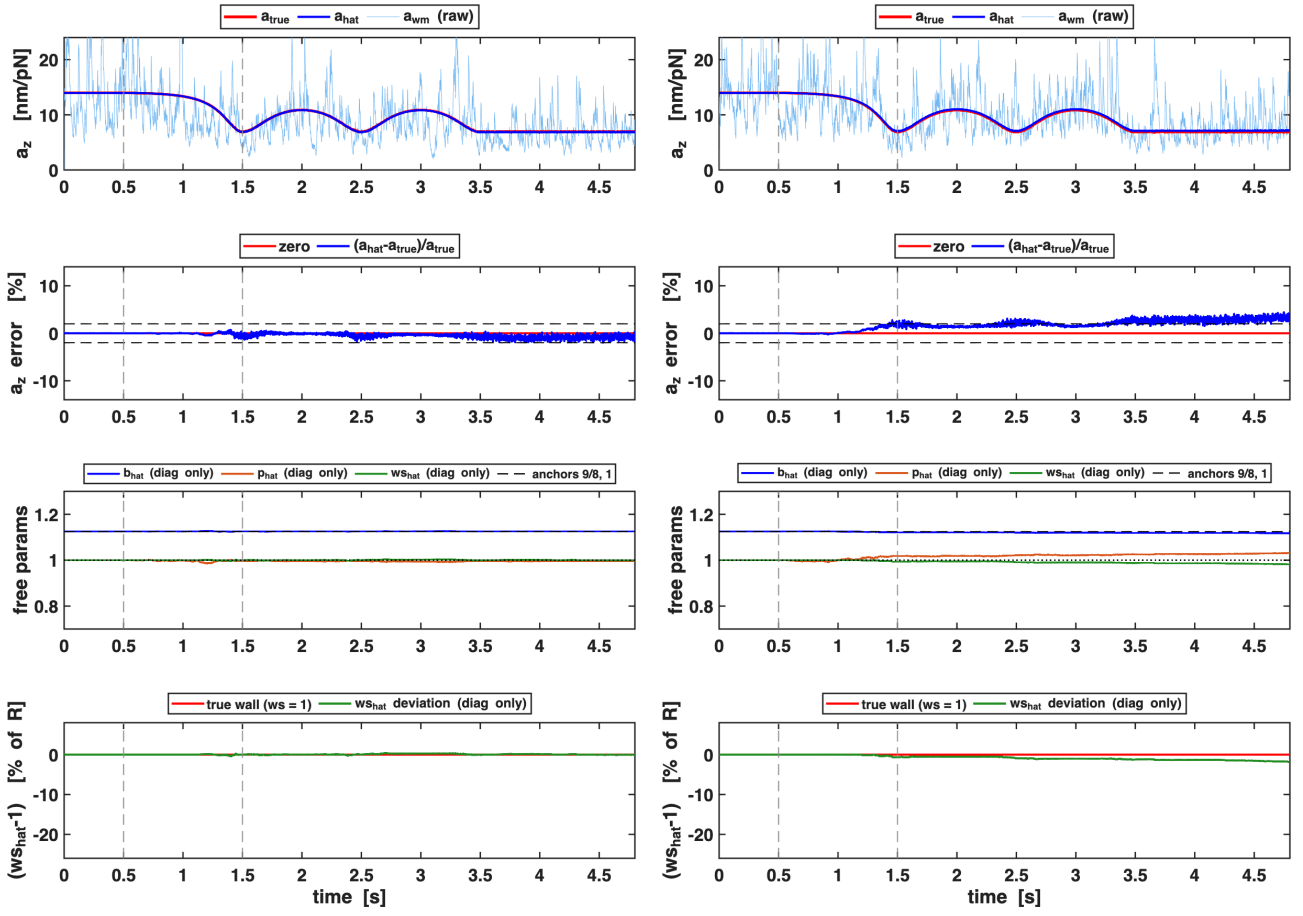
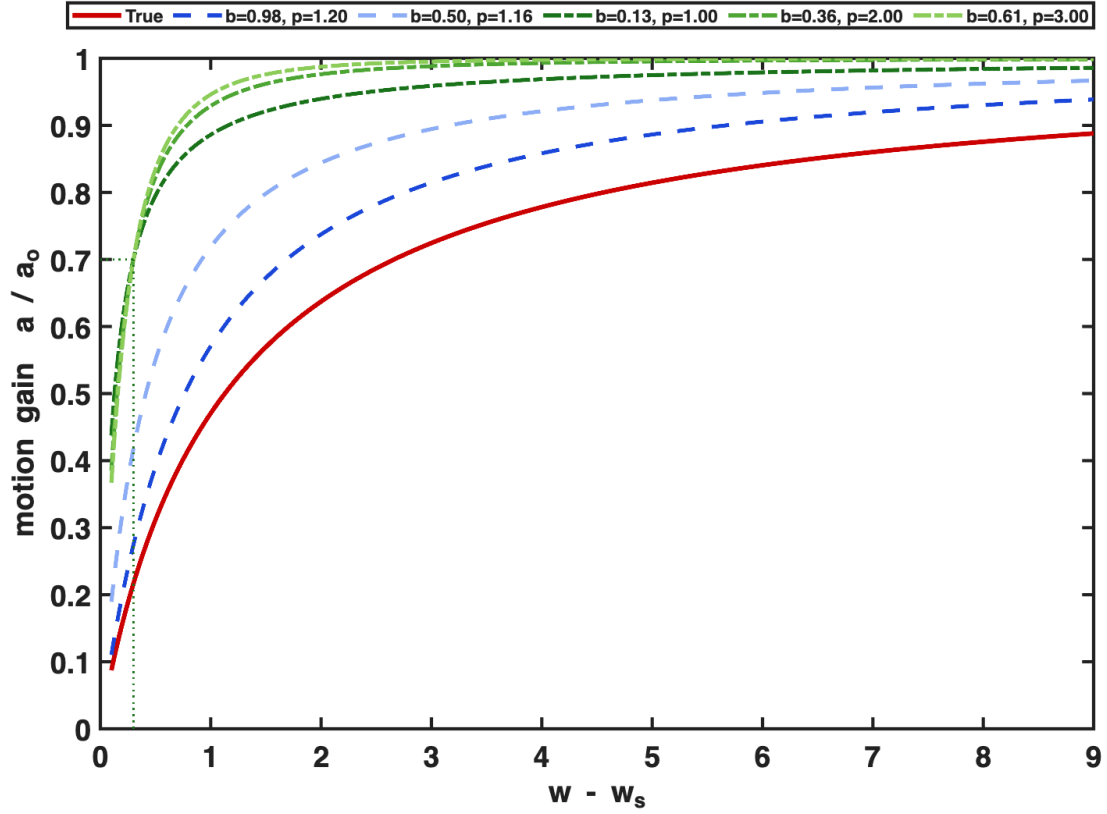


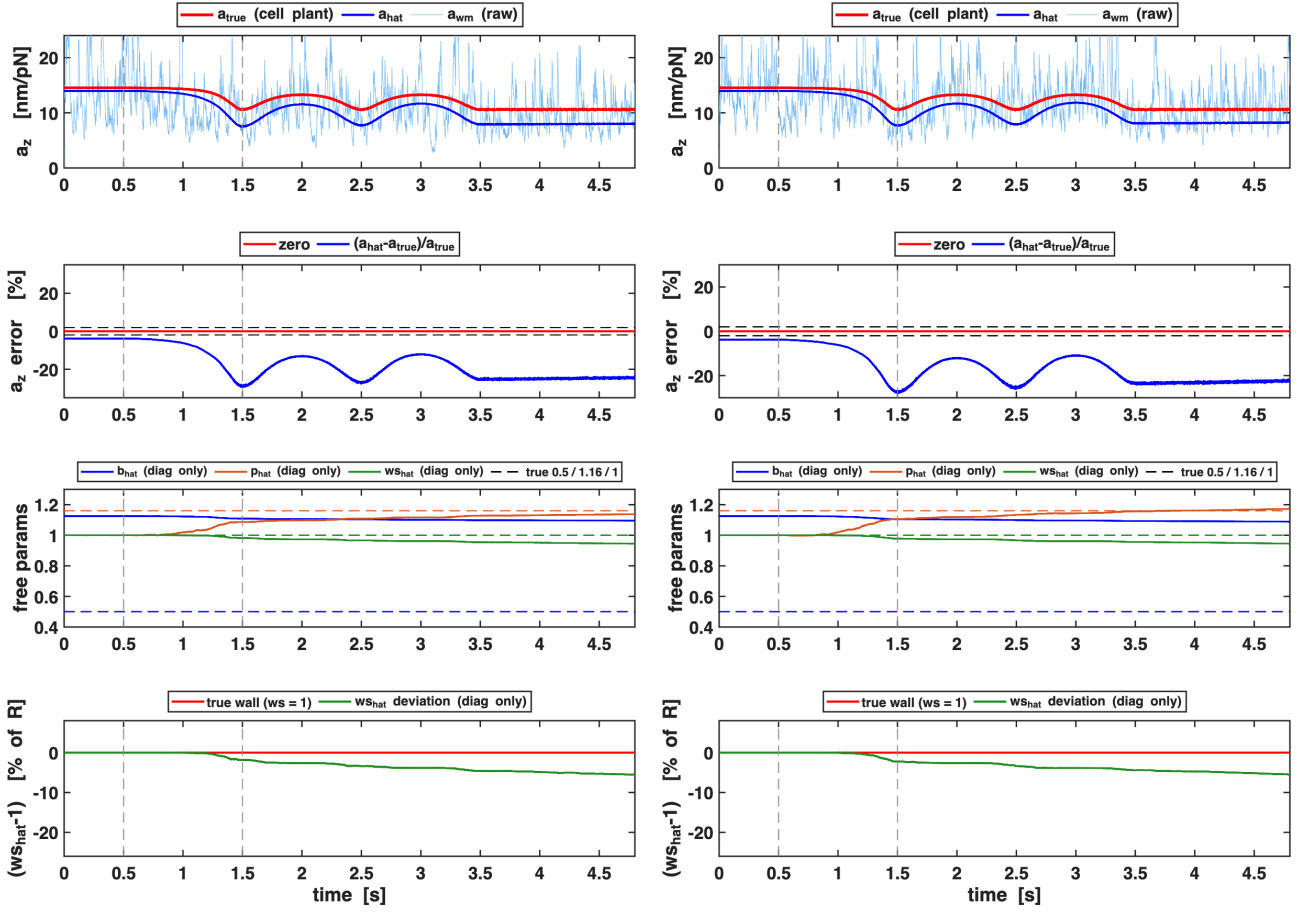
## Selected figures (2026-08-03)



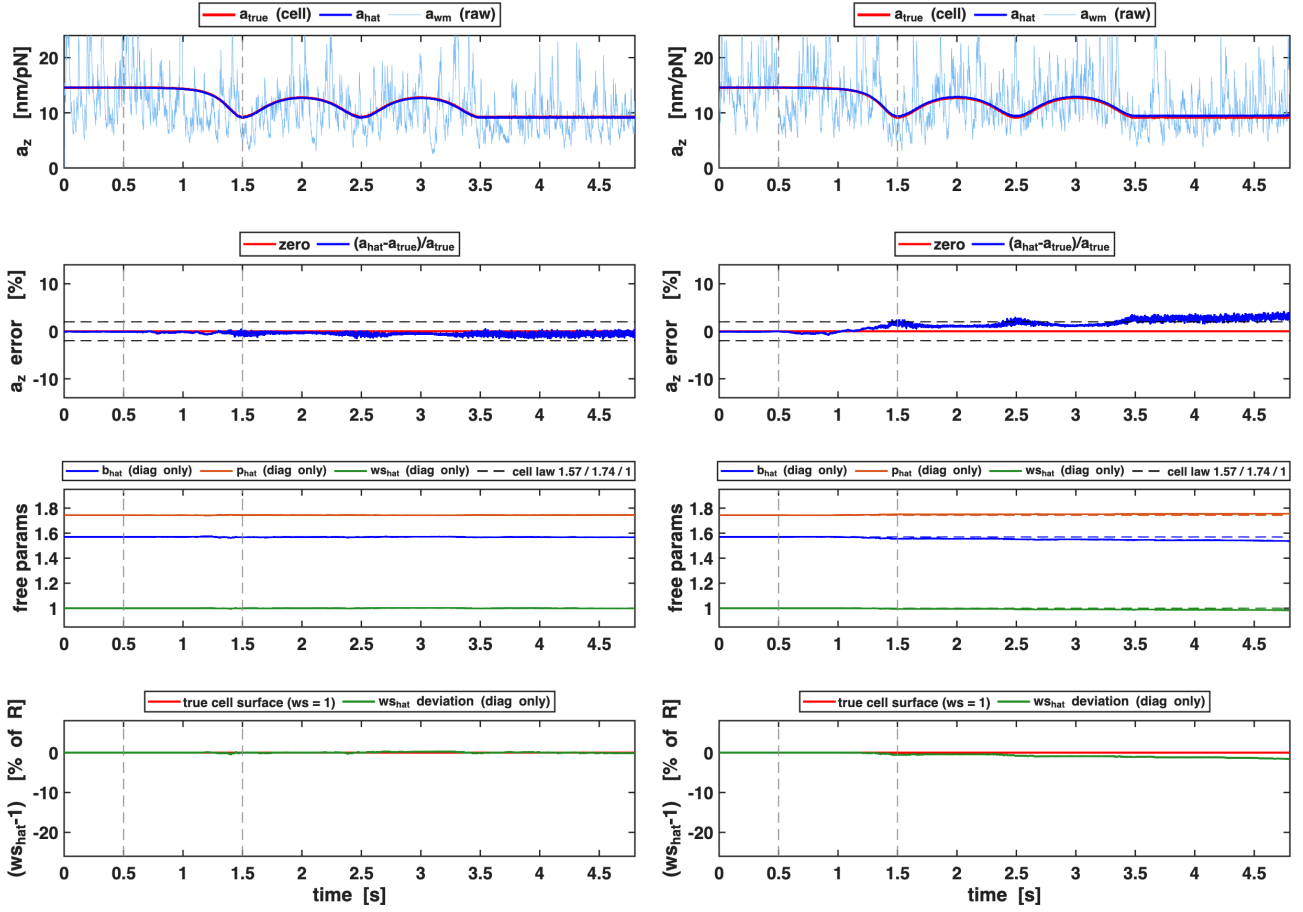
Config 3 — estimate  $(b, p, w_s)$  with the calibrated-grade prior  $\sqrt{P_{w_s}}[0] = 0.028$  (house 0.111 elsewhere unchanged): seed 7 (left) vs seed 11 (right); the seed-11  $\hat{w}_s$  end deviation shrinks to  $-1.8\%$  and the residual realization exits through the  $b, p$  channels.



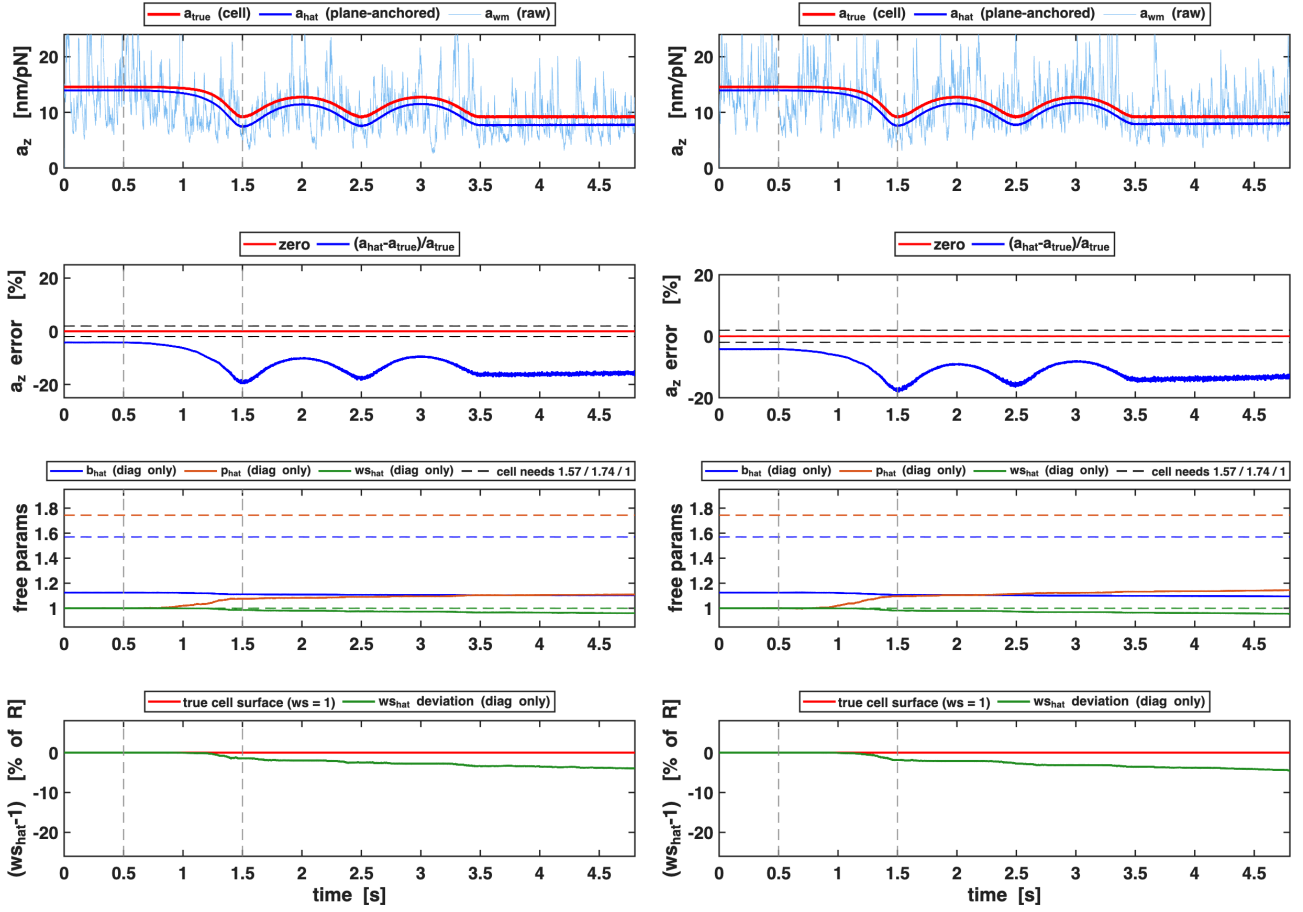
Form B refit to two cell (two-sphere) truths against the rigid-plane truth (red): the family absorbs boundary curvature, but the fitted  $(b, p)$  leave the plane anchors  $(9/8, 1)$ . The three green dash-dot curves are added by hand: all pass through  $\bar{a} = 0.70$  at a gap of  $0.3 R$  (the rigid plane is at  $0.21$  there) with the origin still on the surface, and they show the one-parameter freedom a single point leaves —  $p$  sets the far-field decay  $1 - \bar{a} \sim g^{-p}$  and  $b$  follows as  $b = rG/(1 - r)$ ,  $r = 0.3^{1/p}$ .



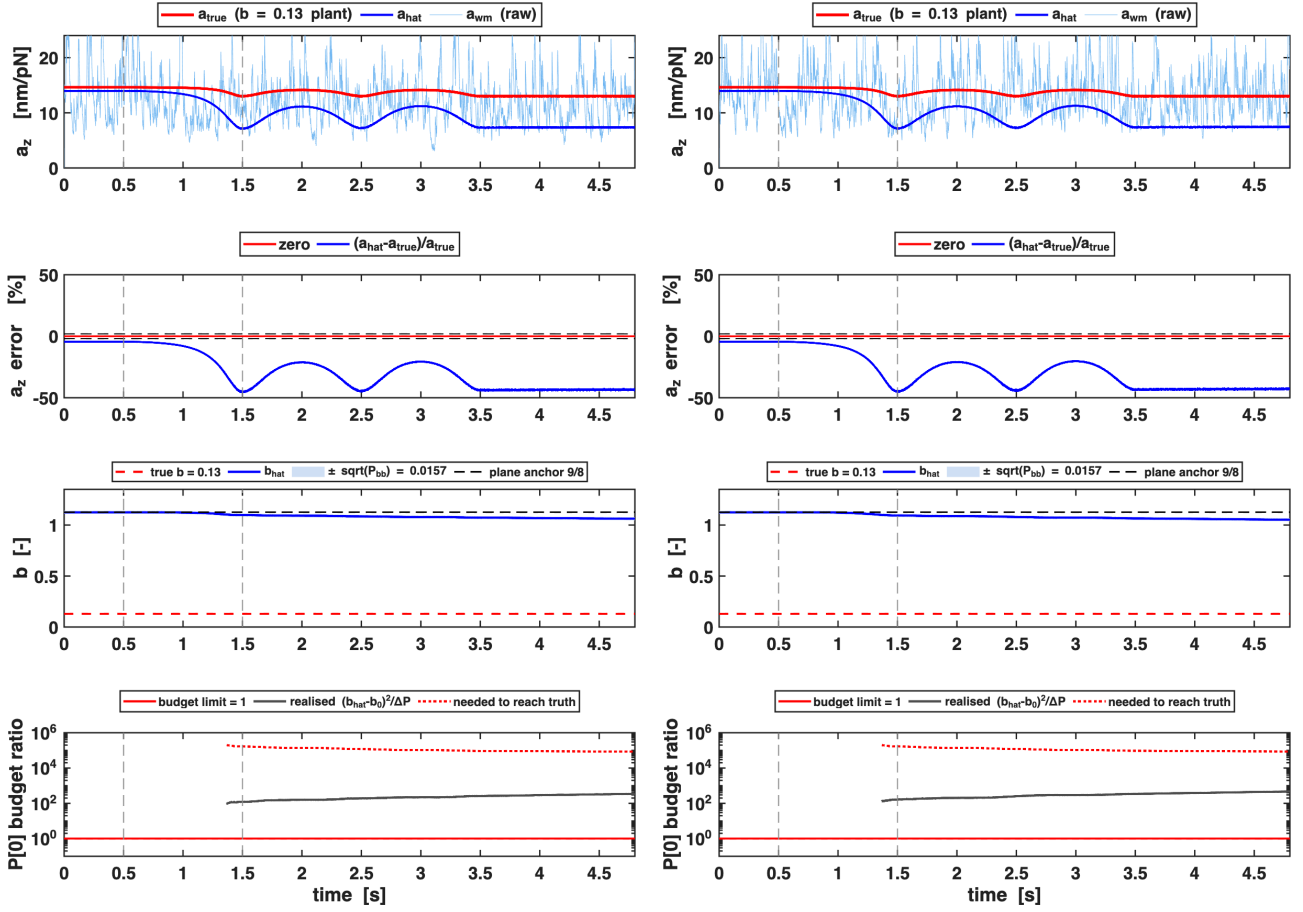
Curve-mismatch arm — same config 3 / prior 0.028 setup, but the TRUE  $z$  curve is the cell fit  $(b, p, w_s) = (0.5, 1.16, 1)$  while the controller keeps every plane-derived prior: gain error blows up (descent peak  $\approx 28\%$ , hold  $\approx -24\%$ ), the  $b, p$  traversal budgets fail by  $\sim 50\times$ , and  $\hat{w}_s$  is dragged  $-5.5\%$  against its own  $2.8\%$  prior — the plane-anchored prior widths cannot absorb a boundary this far outside the family they were derived for.



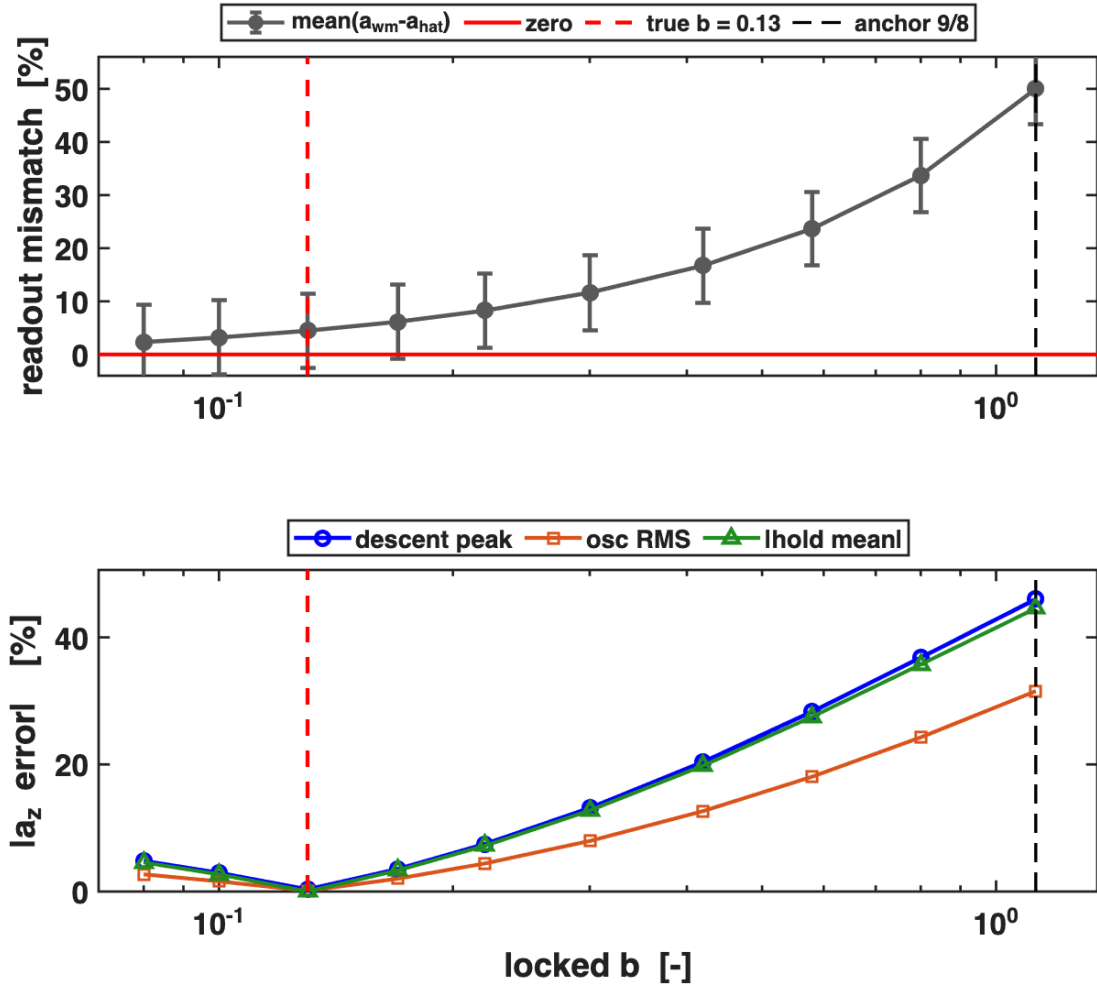
Cell boundary done properly — plant is the exact two-sphere truth for  $R_c = 5 \mu\text{m}$  and the controller carries the cell law  $(b, p, w_{s0}) = (1.570, 1.744, 0.809)$  with widths  $[0.042, 0.022, 0.015]$ , fitted offline to that same published curve (sup 0.086 %): performance returns to the plane baseline (descent 1.10/2.33 %, hold  $-0.61/+2.82$  %,  $\hat{w}_s$  ending  $-0.1/-1.6$  %), so the difficulty was never the curvature but the anchor. This page is the capability bound (“if we knew”), not the operating case.



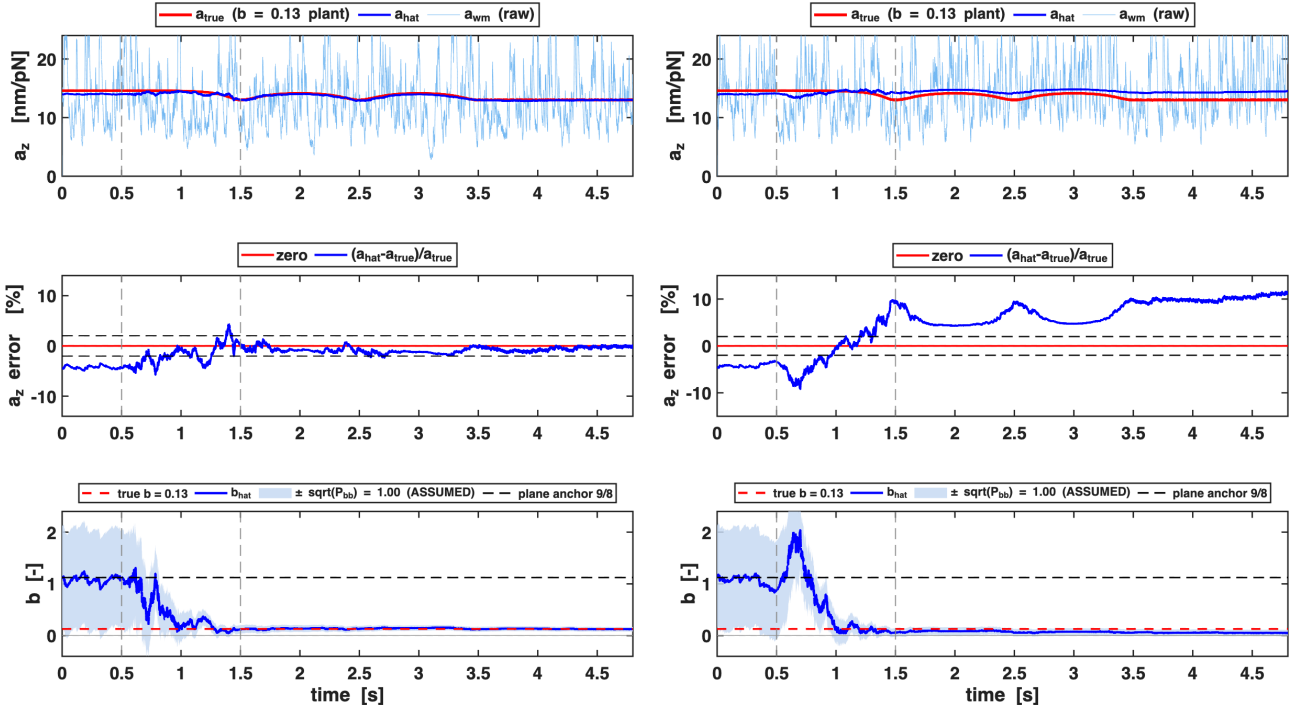
The operating case — same  $R_c = 5 \mu\text{m}$  cell plant, but the controller knows only the plane (anchors 9/8, 1, priors 0.0157/0.0245, origin at the nominal contact): descent peak 19.6/17.8 %, hold  $-16.0 / -13.6$  %,  $b, p$  stall near their plane anchors (final 1.10/1.11 against the 1.570/1.744 the cell demands),  $\hat{w}_s$  absorbs part of the mismatch by drifting  $-4.0 / -4.5$  %, and the traversal budgets fail by  $25\text{--}40\times$  — a plane-anchored filter meeting a cell is wrong by  $\sim 15$  % and says so, but cannot repair itself.



Single-free-parameter arm (2026-08-05) — the TRUE  $z$  curve is the  $p = 1$  member of the page-2 constructed family,  $(b, p, w_s) = (0.13, 1, 1)$ , and the controller frees ONLY  $b$ :  $p$  and  $w_s$  are locked *at their true values*. Nothing is left to blame — no representation error (the truth is a member of the controller’s own family), no  $b$ – $p$  coupling, no  $b$ – $w_s$  degeneracy — so the only remaining obstacle is the prior width itself. The envelope-derived  $\sqrt{P_{bb}}[0] = 0.0157$  places the truth 63.5 prior widths from the 9/8 seed;  $\hat{b}$  moves 0.064/0.073 of the 0.995 it would need and stalls at 1.061/1.052, the traversal budget realises 348/472 against a limit of 1 while reaching the truth would require  $\sim 8.5 \times 10^4$ , and the gain error settles at descent peak 45.5/45.2 %, hold  $-43.5/-43.0$  %. The plant here is a *constructed* test curve, not a fitted cell (the physical cell fits in this document sit at  $b = 0.50$ – $1.57$ , depending on whether the law origin is held on the surface): the page therefore indicts the prior structure, not the boundary physics.



Locked- $b$  profile for the same constructed plant —  $b$  is LOCKED on a swept grid instead of estimated, so the EKF prior plays no part. Bottom: the truth-referenced gain error is a clean V bottoming *exactly* at  $b = 0.13$  (descent 0.33 %, osc 0.12 %, hold +0.01 % — the plane baseline), which verifies the Form B machinery: locked at the right constant the filter is exact, and every percent above that floor is the constant being wrong. Top: the data-only statistic — the mean readout-versus-model mismatch, computable without knowing the truth — climbs monotonically across the grid, +2.3 % at  $b = 0.08$  to  $+50.0 \pm 6.7$  % at the plane anchor, where it stands  $7.5\sigma$  from zero on two seeds. The run therefore *does* carry, in a truth-free statistic, the information that the anchored  $b$  is wrong; the companion page’s filter simply cannot act on it, because its prior places the truth 63.5 widths away. Identifiability is not the obstacle — the prior width is. Two caveats on reading  $b$  off this curve: the mismatch is monotonic with no feature at the truth (locating  $b$  needs the readout offset characterised first — it is  $+4.5 \pm 7.0$  % there, consistent with zero but unresolved at  $N = 2$ ), and near the truth the slope is only  $\sim 6$  % per e-fold against that 7 % scatter, so two seeds fix  $b$  to a factor of a few, not to a percent. The filter’s own whitened  $y_2$  channel is a poor route to the same information: its innovation mean tracks  $b$  over  $12\times$ , but the per-sample RMS is noise-dominated and bottoms at  $b = 0.80$ .



Widened- $b$ -prior arm — same plant and same locks as two pages back, but  $\sqrt{P_{bb}}[0]$  is opened up so the truth is inside the prior. **The widths are ASSUMED, not derived** (provisional, to bracket the travel  $|9/8 - 0.13| = 0.995$ ); a committed width must come from a declared boundary family.

Ladder,  $\sqrt{P_{bb}} \rightarrow \hat{b}_{\text{final}}$  (seed 7 / 11), descent error, budget:  $0.0157 \rightarrow 1.061/1.052$ , 45.5/45.2 %, 348/472;  $0.10 \rightarrow 0.316/0.185$ , 29.3/24.8 %, 96/136;  $0.25 \rightarrow 0.132/0.051$ , 8.5/6.9 %, 16.5/19.7;  $0.50 \rightarrow 0.118/0.051$ , 5.5/9.6 %, 4.1/4.7;  $1.00 \rightarrow 0.123/0.051$ , 5.7/9.8 %, 1.01/1.16. Four readings. (i)

Once the prior stops forbidding it the filter *does* walk  $b$  to the truth, and it does so during the descent, exactly where the wall-gap information lives. (ii) The  $P[0]$  budget criterion turns out to mean literally what it says — to move a distance  $\Delta$  lawfully the prior must satisfy  $\sqrt{P[0]} \geq \Delta$ , and at  $\sqrt{P_{bb}} = 1.00 \approx \Delta$  the realised ratio lands at 1.01/1.16, on the limit. (iii) The prior stops binding above  $\sqrt{P_{bb}} \approx 0.25$ : from there the answer is data-limited, not prior-limited, and widening further buys nothing. (iv) What data-limited costs:  $\hat{b}$  scatters 0.123 vs 0.051 across two seeds (factor 2.4) and the hold error splits  $-0.5\%$  vs  $+10.2\%$ , so the gain recovers to the few-percent level but not to the 0.33 % floor that *knowing*  $b$  gives. Caveat on the container: a linear Gaussian prior with mean  $9/8$  and  $\sqrt{P} = 1.0$  puts  $\sim 13\%$  of its mass at  $b < 0$ , which is unphysical for a length — a genuinely wide prior on  $b$  probably wants a log parametrisation, which is a state-level change and was not attempted.